

A2 Warm Up for Quiz 8.3

1. For which values of x is the expression undefined?

$$\frac{x^2 - 4}{x + 1}$$

$$x + 1 = 0$$

$$x = -1$$

2. Find all vertical asymptotes of the following function.

$$f(x) = \frac{3x^2 - 18x - 81}{x^2 - 4x - 32}$$

$$3(x^2 - 6x - 27) - 9$$

$$x^2 - 4x - 32$$

$$3(x - 9)(x + 3)$$

$$(x - 8)(x + 4)$$

$$x - 8 = 0$$

$$x = 8$$

$$x + 4 = 0$$

$$x = -4$$

3. Find all horizontal asymptotes of the following function.

$$f(x) = \frac{x + 4}{2(5x + 4)(x + 4)}$$

$$\frac{x}{10x^2} \rightarrow \text{Bottom Heavy}$$

$$y = 0$$

Name _____ Per _____

4. Find all x -intercepts of the following function. Write your answer or answers as coordinate points.

$$f(x) = \frac{x^2 - 10x + 24}{x + 9}$$

$$(x - 6)(x - 4)$$

$$x + 9$$

$$x - 6 = 0$$

$$x = 6$$

$$x - 4 = 0$$

$$x = 4$$

$$(6, 0)$$

$$(4, 0)$$

5. Find all y -intercepts of the following function. Express your answer as a coordinate point in simplest form.

$$f(x) = \frac{2x^2 - 5x + 2}{3x - 6}$$

$$= \frac{2(0)^2 - 5(0) + 2}{3(0) - 6}$$

$$= \frac{-2}{6} = -\frac{1}{3}$$

$$(0, -\frac{1}{3})$$

6. Determine each feature of the graph of the function $f(x) = \frac{-3x - 6}{x^2 - 4}$, if it exists. If the feature does not exist, leave it blank.

Horizontal Asymptote: $y = 0$ y -Intercept: $(0, \frac{3}{2})$

Vertical Asymptote: $x = 2$ x -Intercept: $(\frac{3}{4}, 0)$

Hole: $(-2, \frac{3}{4})$

$$\frac{-3(x + 2)}{(x - 2)(x + 2)}$$

$$\frac{-3}{x - 2}$$

$$HA: \frac{-3}{x}$$

$$\text{Bottom Heavy}$$

$$y = 0$$

$$\text{Hole}$$

$$x + 2 = 0$$

$$x = -2$$

$$\frac{-3}{-2 - 2} = \frac{-3}{-4} = \frac{3}{4}$$

$$(-2, \frac{3}{4})$$

$$y\text{-int } \frac{-3}{x - 2}$$

$$= \frac{-3}{0 - 2}$$

$$= \frac{3}{2}$$

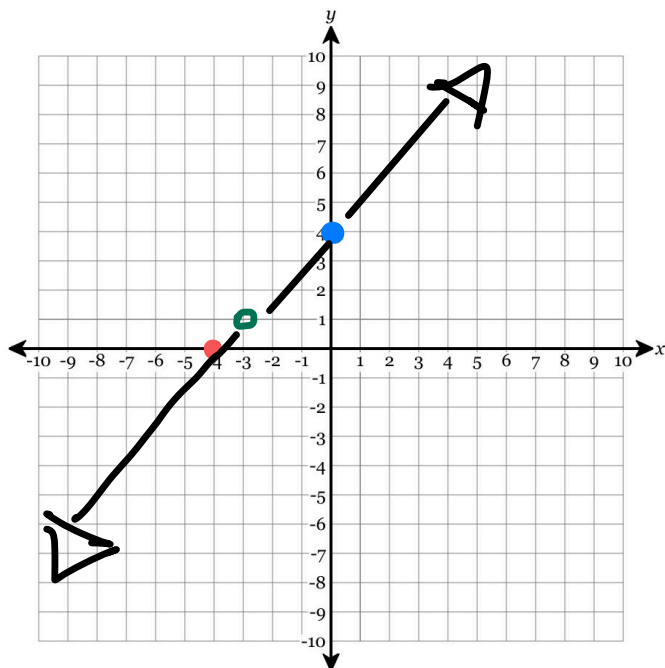
$$VA: x - 2 = 0$$

$$x = 2$$

$$3 \neq 0 \text{ NO } x\text{-int}$$

7. Identify all vertical and horizontal asymptotes, x- and y-intercepts, and holes, and then use them to graph the rational function below.

$$f(x) = \frac{x^2 + 7x + 12}{x + 3}$$



$$f(x) = \frac{(x+4)(x+2)}{x+3}$$

$$x+3$$

$$f(x) = x+4$$

$$\text{hole: } x+3=0 \\ x=-3$$

$$-3+4 = 1$$

$$(-3, 1)$$

$$4A \quad \frac{x}{1}$$

None

for Hany

$$x+4=0$$

$$x+4=0$$

$$(-4, 0)$$

$$y\text{-int}$$

$$0+4=4$$

$$0, 4$$

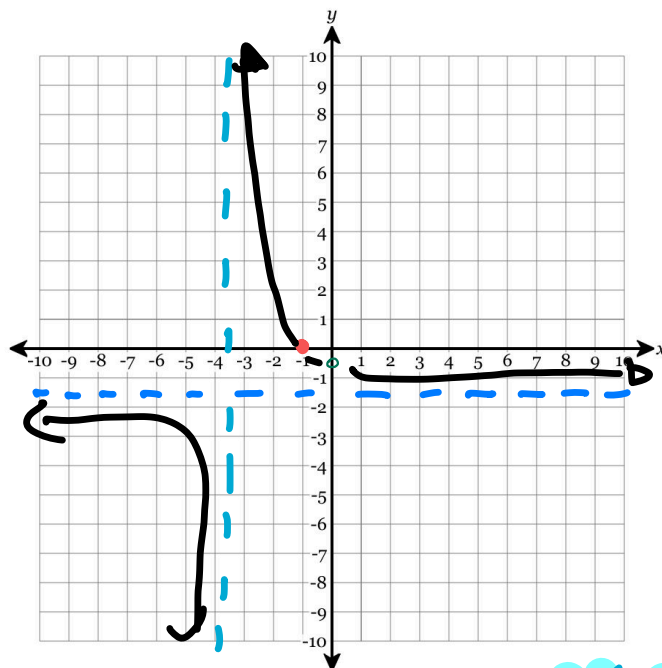
$$VA$$

$$0 \neq 1$$

none

8. Identify all vertical and horizontal asymptotes, x- and y-intercepts, and holes, and then use them to graph the rational function below.

$$f(x) = \frac{-3x^2 - 3x}{2x^2 + 7x}$$



$$f(x) = \frac{-3x(x+1)}{2x+7} = \frac{-3(x+1)}{2x+7}$$

$$\text{Hole: } x=0$$

$$f(0) = \frac{-3(0+1)}{2(0)+7}$$

$$= -\frac{3}{7}$$

$$(0, -\frac{3}{7})$$

$$x\text{-int}$$

$$x+1=0$$

$$x=-1$$

$$(-1, 0)$$

$$HA: \frac{-3x}{2x} = -\frac{3}{2}$$

$$y = -\frac{3}{2}$$

$$y\text{-int} \quad \frac{-3(0+1)}{2(0)+7} = -\frac{3}{7}$$

No y-int b/c hole is there (0, -3/7)

$$VA:$$

$$2x+7=0$$

$$x = -\frac{7}{2}$$

$$x = -3.5$$

$$x^2 + 10x + 24$$

$$a = 1$$

$$b = 10$$

$$c = 24$$

$$\begin{array}{c} a \cdot c \\ 24 \\ \hline \begin{array}{cc} 4 & 6 \end{array} \\ \hline 10 \\ b \end{array}$$

$$\begin{array}{r} 24 \\ \hline 3 \cdot 8 \\ \hline 4 \cdot 6 \end{array}$$

$$(x + 4)(x + 6)$$

$$x^2 - 11x + 24$$

$$a = 1$$

$$b = -11$$

$$c = 24$$

$$\begin{array}{c} a \cdot c \\ 24 \\ \hline \begin{array}{cc} -3 & -8 \end{array} \\ \hline -11 \\ b \end{array}$$

$$\begin{array}{r} 24 \\ \hline \end{array}$$

$$- 2 \cdot 12$$

$$- 3 \cdot 8$$

$$(x - 3)(x - 8)$$

$$x^2 + 2x - 24$$

$$a = 1$$

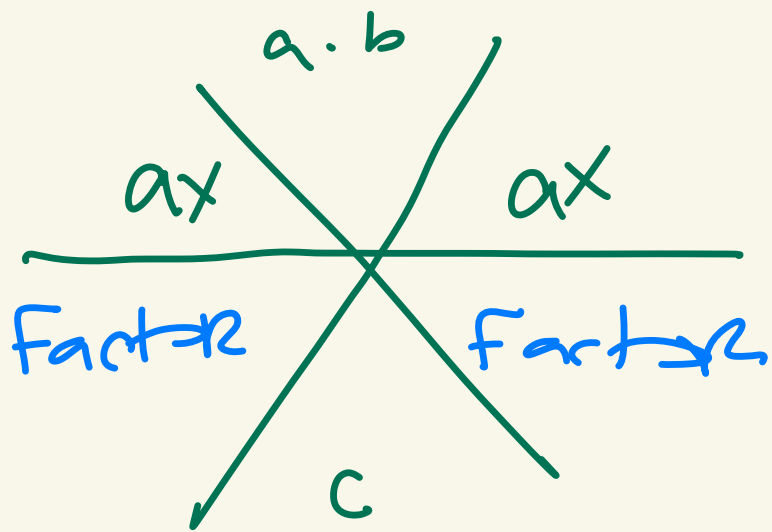
$$b = 2$$

$$c = -24$$

$$\begin{array}{r} -24 \\ -4 \quad 6 \\ \hline 2 \end{array}$$

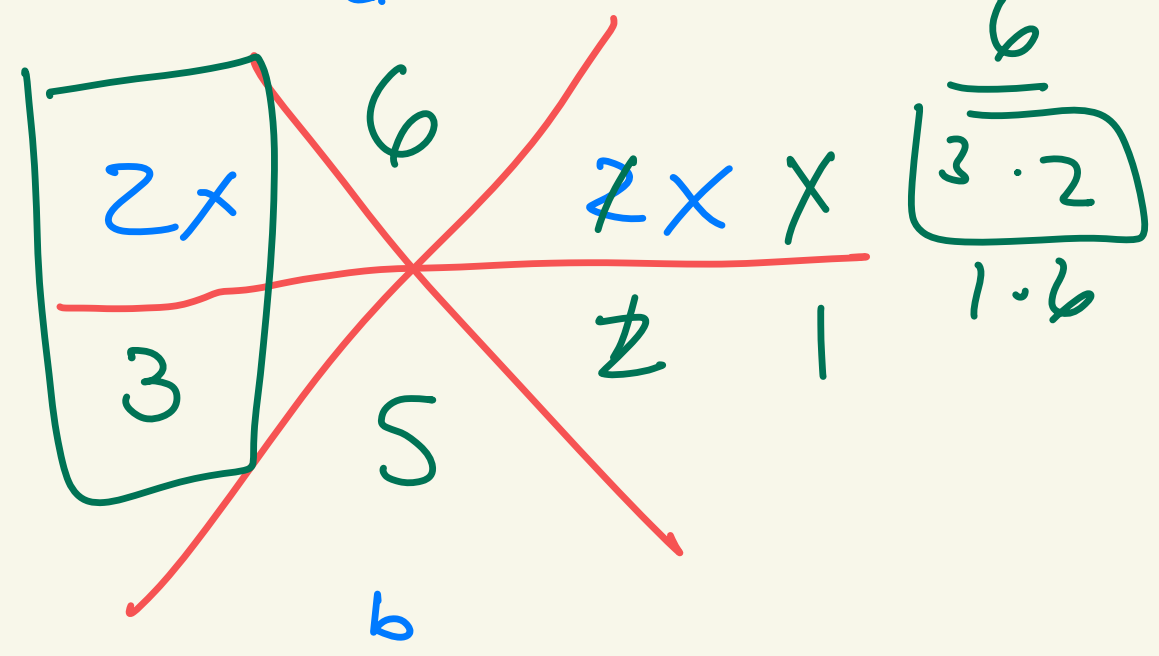
$$(x - 4)(x + 6)$$

$$\begin{array}{r} 24 \\ \hline 8 \cdot 3 \\ \boxed{-4 \cdot 6} \\ 2 \cdot 12 \\ 1 \cdot 24 \end{array}$$



$(2x^2 + 5x + 3)$ $a \cdot c$

$a = 2$
 $b = 5$
 $c = 3$



$(2x + 3)(x + 1)$

$$3p^2 - 2p - 5$$

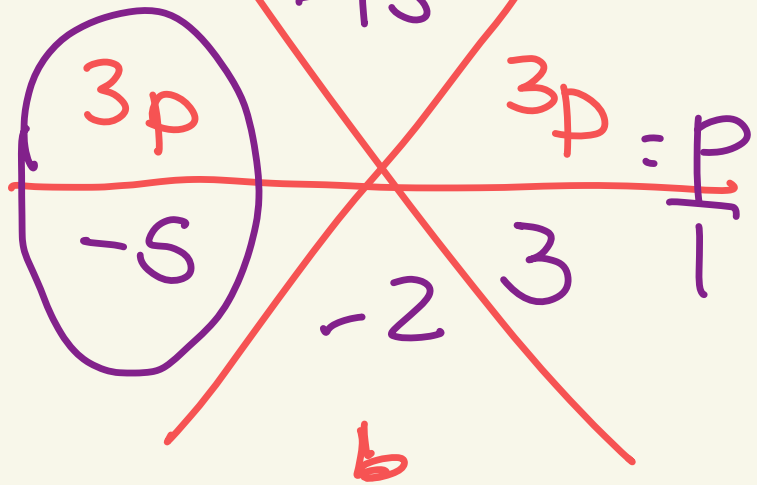
$$a = 3$$

$$\frac{-15}{3} = -5$$

$$3 \div 5$$

$$b = -2$$

$$c = -5$$



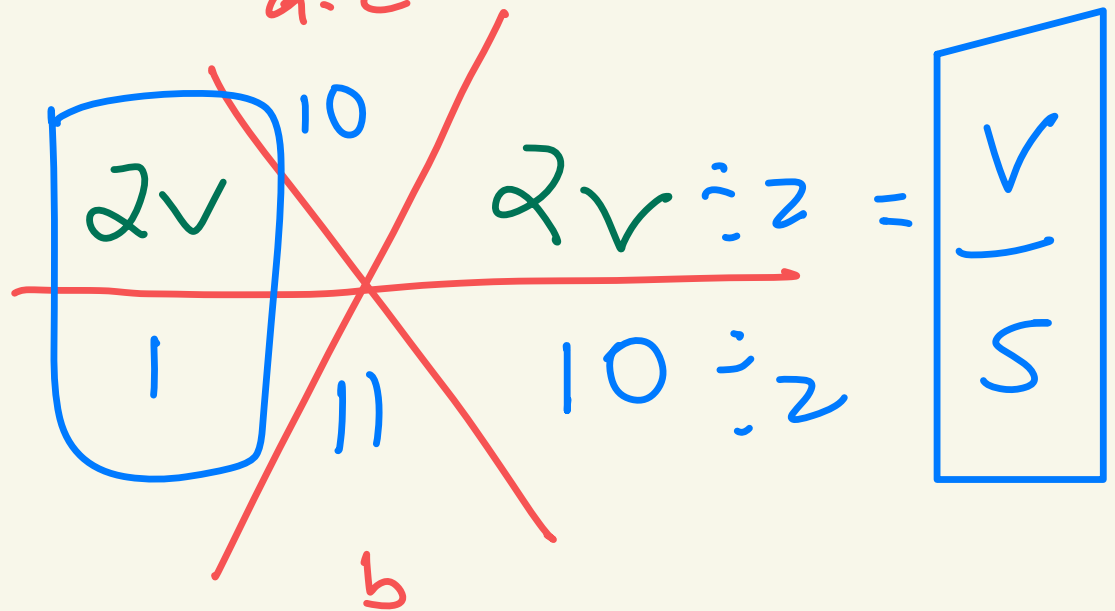
$$(3p - 5)(p + 1)$$

$$2v^2 + 11v + 5$$

$$a = 2$$

$$b = 11$$

$$c = 5$$



$$2v \div 2 = v$$

$$10 \div 2 = 5$$

$$\frac{v}{5}$$

$$(2v + 1)(v + 5)$$

$$2n^2 + 5n + 2$$

$$a = 2$$

$$b = 5$$

$$c = 2$$

$$\begin{array}{cc}
 & a-c \\
 & 4 \\
 \hline
 2n & 2n \\
 \hline
 1 & 4 \\
 & b \\
 & 1
 \end{array}$$

$$\begin{array}{c}
 \sqrt{} \\
 \hline
 n \\
 \hline
 2
 \end{array}$$

$$3n^2 - 8n + 4$$